

D:\Data Khoa luan cua Thai\CB-SEM test 1.amw

Analysis Summary

Date and Time

Date: Saturday, September 12, 2026
Time: 8:51:57 PM

Title

Cb-sem test 1: Saturday, September 12, 2026 8:51 PM

Groups

Group number 1 (Group number 1)

Notes for Group (Group number 1)

The model is recursive.
Sample size = 240

Variable Summary (Group number 1)

Your model contains the following variables (Group number 1)

Observed, endogenous variables

- TE1
- TE2
- TE3
- TE4
- TE5
- TE6
- TE7
- TE8
- KT1
- KT2
- KT3
- KT4
- KT5
- KT6
- KT7
- KA1
- KA2
- KA3
- KA4
- KA6
- JP1
- JP2
- JP3
- JP4
- JP5

Unobserved, endogenous variables

- KT
- KA
- JP

Unobserved, exogenous variables

- e1

e25
e27
e28
e26

Variable counts (Group number 1)

Number of variables in your model: 57
Number of observed variables: 25
Number of unobserved variables: 32
Number of exogenous variables: 29
Number of endogenous variables: 28

Parameter Summary (Group number 1)

	Weights	Covariances	Variances	Means	Intercepts	Total
Fixed	32	0	0	0	0	32
Labeled	0	0	0	0	0	0
Unlabeled	27	0	29	0	0	56
Total	59	0	29	0	0	88

Models

Default model (Default model)

Notes for Model (Default model)

Computation of degrees of freedom (Default model)

Number of distinct sample moments: 325
Number of distinct parameters to be estimated: 56
Degrees of freedom (325 - 56): 269

Result (Default model)

Minimum was achieved
Chi-square = 381.537
Degrees of freedom = 269
Probability level = .000

Group number 1 (Group number 1 - Default model)

Estimates (Group number 1 - Default model)

Scalar Estimates (Group number 1 - Default model)

Maximum Likelihood Estimates

Regression Weights: (Group number 1 - Default model)

	Estimate	S.E.	C.R.	P	Label
KA <--- TE	.644	.082	7.839	***	
KT <--- TE	.513	.105	4.902	***	
KT <--- KA	.294	.106	2.760	.006	
JP <--- TE	.449	.101	4.444	***	
JP <--- KT	.216	.075	2.872	.004	
JP <--- KA	.441	.104	4.246	***	
TE1 <--- TE	1.000				
TE2 <--- TE	1.043	.090	11.601	***	

TE7	<---	TE	1.114	.089	12.455	***
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Standardized Regression Weights: (Group number 1 - Default model)

	Estimate
KA <--- TE	.669
KT <--- TE	.459
KT <--- KA	.253
JP <--- TE	.399
JP <--- KT	.214
JP <--- KA	.377
TE1 <--- TE	.763
TE2 <--- TE	.731
TE3 <--- TE	.722
TE4 <--- TE	.764
TE5 <--- TE	.705
TE6 <--- TE	.707
TE8 <--- TE	.696
KT1 <--- KT	.719
KT2 <--- KT	.743
KT3 <--- KT	.729
KT4 <--- KT	.670
KT5 <--- KT	.820
KT6 <--- KT	.743
KT7 <--- KT	.760
KA1 <--- KA	.674
KA2 <--- KA	.735
KA3 <--- KA	.659
KA4 <--- KA	.740
KA6 <--- KA	.734
JP1 <--- JP	.702
JP2 <--- JP	.634
JP3 <--- JP	.654
JP4 <--- JP	.638
JP5 <--- JP	.661
TE7 <--- TE	.778

Variances: (Group number 1 - Default model)

	Estimate	S.E.	C.R.	P	Label
TE	.320	.047	6.783	***	
e26	.164	.032	5.129	***	
e28	.228	.039	5.890	***	
e27	.103	.025	4.092	***	
e1	.229	.024	9.517	***	
e2	.303	.031	9.767	***	
e3	.321	.033	9.830	***	
e4	.294	.031	9.512	***	
e5	.336	.034	9.934	***	
e6	.360	.036	9.920	***	
e7	.259	.028	9.376	***	
e8	.352	.035	9.983	***	
e9	.373	.038	9.711	***	
e10	.245	.026	9.527	***	
e11	.275	.029	9.639	***	

JP4	.407
JP3	.427
JP2	.402
JP1	.493
KA6	.539
KA4	.548
KA3	.435
KA2	.540
KA1	.454
KT7	.577
KT6	.552
KT5	.672
KT4	.448
KT3	.532
KT2	.552
KT1	.518
TE8	.484
TE7	.605
TE6	.500
TE5	.497
TE4	.583
TE3	.521
TE2	.535
TE1	.583

Matrices (Group number 1 - Default model)

Residual Covariances (Group number 1 - Default model)

	JP5	JP4	JP3	JP2	JP1	KA6	KA4	KA3	KA2	KA1	KT7	KT6	KT5	KT4	K'
JP5	.000														
JP4	.031	.000													
JP3	-.007	.019	.000												
JP2	-.010	.005	-.011	.000											
JP1	-.008	-.019	.007	.001	.000										
KA6	.026	.004	.029	.005	.043	.000									
KA4	-.009	.011	-.002	.017	-.009	-.013	.000								
KA3	.018	-.043	-.056	-.021	-.016	.011	-.002	.000							
KA2	-.025	-.074	-.003	.026	.014	-.014	-.005	.039	.000						
KA1	.005	-.010	.009	-.011	.007	.006	.005	-.005	-.010	.000					
KT7	.028	.053	.027	.017	-.039	.006	.026	-.041	.028	-.033	.000				
KT6	.001	-.004	.108	-.008	-.027	.009	-.032	-.077	.045	-.013	.033	.000			
KT5	.037	.053	.066	.009	-.001	.038	.024	.021	.070	-.017	.006	.016	.000		
KT4	-.064	-.032	-.008	-.004	-.066	-.006	-.003	.069	.087	-.071	-.037	-.043	.009	.000	
KT3	-.051	.046	.007	-.066	-.046	-.048	-.007	-.041	.030	-.086	.000	.014	-.037	.032	.0
KT2	-.022	.013	-.014	-.035	-.064	-.046	-.036	-.045	-.019	-.009	.005	.017	-.039	.007	.0
KT1	.004	.034	.036	.030	-.011	.047	.055	.007	.058	-.039	-.023	-.071	.036	.038	-.0
TE8	-.004	-.048	-.015	.001	-.013	.020	.011	-.036	.022	.017	-.022	-.045	-.003	-.054	-.0
TE7	-.006	-.015	-.019	-.005	.010	.011	.047	-.036	.031	.019	-.011	-.021	.026	.051	.0
TE6	-.055	.012	-.072	-.026	.017	.019	.073	-.011	.041	.039	-.014	.004	.029	.003	-.0
TE5	-.018	.009	-.021	.020	-.007	.002	.055	.021	-.005	.024	.056	.057	.037	.056	.0
TE4	-.008	-.062	-.012	.037	.064	-.005	.007	-.062	-.018	-.019	.039	-.022	.020	.028	-.0
TE3	-.015	-.011	.017	.020	.013	-.027	.034	-.077	.003	.052	-.050	.013	.019	.006	-.0
TE2	.008	-.050	-.021	-.002	.024	-.025	-.024	-.103	-.056	-.003	-.038	-.004	-.050	-.024	-.0
TE1	.071	.017	-.030	.030	.046	-.029	.003	-.030	-.032	.012	-.027	-.007	.026	.001	-.0

TE7	-.126	-.269	-.366	-.111	.198	.230	.953	-.700	.603	.420	-.214	-.421	.501
TE6	-1.105	.211	-1.313	-.500	.319	.391	1.428	-.216	.763	.830	-.270	.067	.525
TE5	-.377	.158	-.405	.399	-.137	.033	1.127	.406	-.100	.526	1.125	1.122	.712
TE4	-.157	-1.093	-.228	.727	1.200	-.106	.135	-1.166	-.332	-.416	.754	-.422	.373
TE3	-.314	-.205	.319	.398	.255	-.559	.690	-1.510	.053	1.166	-1.006	.249	.353
TE2	.169	-.908	-.399	-.042	.469	-.526	-.491	-2.042	-1.093	-.072	-.767	-.086	-.957
TE1	1.622	.335	-.622	.673	.977	-.670	.069	-.651	-.674	.286	-.588	-.141	.554

Bootstrap (Group number 1 - Default model)

Bootstrap standard errors (Group number 1 - Default model)

Scalar Estimates (Group number 1 - Default model)

Regression Weights: (Group number 1 - Default model)

Parameter	SE	SE-SE	Mean	Bias	SE-Bias
KA <--- TE	.092	.001	.648	.004	.001
KT <--- TE	.125	.001	.521	.009	.002
KT <--- KA	.131	.001	.295	.001	.002
JP <--- TE	.131	.001	.463	.014	.002
JP <--- KT	.087	.001	.216	.000	.001
JP <--- KA	.127	.001	.436	-.005	.002
TE1 <--- TE	.000	.000	1.000	.000	.000
TE2 <--- TE	.098	.001	1.049	.006	.001
TE3 <--- TE	.104	.001	1.050	.006	.001
TE4 <--- TE	.102	.001	1.139	.006	.001
TE5 <--- TE	.106	.001	1.026	.008	.001
TE6 <--- TE	.120	.001	1.068	.009	.002
TE8 <--- TE	.102	.001	1.022	.006	.001
KT1 <--- KT	.000	.000	1.000	.000	.000
KT2 <--- KT	.081	.001	.874	.005	.001
KT3 <--- KT	.084	.001	.888	.004	.001
KT4 <--- KT	.093	.001	.966	.002	.001
KT5 <--- KT	.106	.001	1.210	.007	.002
KT6 <--- KT	.113	.001	1.080	.009	.002
KT7 <--- KT	.104	.001	1.074	.006	.001
KA1 <--- KA	.000	.000	1.000	.000	.000
KA2 <--- KA	.132	.001	1.252	.005	.002
KA3 <--- KA	.131	.001	1.117	.001	.002
KA4 <--- KA	.129	.001	1.203	.011	.002
KA6 <--- KA	.103	.001	1.154	.006	.001
JP1 <--- JP	.000	.000	1.000	.000	.000
JP2 <--- JP	.092	.001	.880	.005	.001
JP3 <--- JP	.092	.001	.965	.004	.001
JP4 <--- JP	.112	.001	.986	.004	.002
JP5 <--- JP	.091	.001	.886	.005	.001
TE7 <--- TE	.111	.001	1.124	.010	.002

Standardized Regression Weights: (Group number 1 - Default model)

Parameter	SE	SE-SE	Mean	Bias	SE-Bias
KA <--- TE	.046	.000	.669	-.001	.001
KT <--- TE	.089	.001	.462	.003	.001
KT <--- KA	.105	.001	.251	-.002	.001
JP <--- TE	.105	.001	.408	.009	.001

JP1	<---	JP	.044	.000	.702	-.001	.001
JP2	<---	JP	.041	.000	.635	.000	.001
JP3	<---	JP	.044	.000	.652	-.001	.001
JP4	<---	JP	.046	.000	.636	-.002	.001
JP5	<---	JP	.043	.000	.662	.000	.001
TE7	<---	TE	.028	.000	.778	.000	.000

Variances: (Group number 1 - Default model)

Parameter	SE	SE-SE	Mean	Bias	SE-Bias
TE	.052	.001	.319	-.001	.001
e26	.033	.000	.163	-.001	.000
e28	.043	.000	.224	-.004	.001
e27	.024	.000	.099	-.005	.000
e1	.027	.000	.226	-.003	.000
e2	.031	.000	.301	-.002	.000
e3	.032	.000	.317	-.004	.000
e4	.032	.000	.291	-.003	.000
e5	.040	.000	.332	-.004	.001
e6	.047	.000	.356	-.003	.001
e7	.034	.000	.257	-.002	.000
e8	.045	.000	.349	-.003	.001
e9	.050	.000	.369	-.004	.001
e10	.024	.000	.242	-.003	.000
e11	.031	.000	.273	-.002	.000
e12	.064	.001	.455	-.002	.001
e13	.038	.000	.280	-.002	.001
e14	.051	.001	.368	-.003	.001
e15	.041	.000	.331	-.003	.001
e16	.041	.000	.353	-.003	.001
e17	.064	.001	.389	-.003	.001
e18	.078	.001	.476	-.003	.001
e19	.055	.001	.346	-.001	.001
e20	.047	.000	.331	-.003	.001
e21	.059	.001	.415	-.003	.001
e22	.055	.001	.457	-.004	.001
e23	.066	.001	.502	-.002	.001
e24	.071	.001	.568	-.003	.001
e25	.048	.000	.401	-.004	.001

Squared Multiple Correlations: (Group number 1 - Default model)

Parameter	SE	SE-SE	Mean	Bias	SE-Bias
KA	.061	.001	.449	.001	.001
KT	.057	.001	.438	.008	.001
JP	.050	.000	.757	.010	.001
JP5	.056	.001	.440	.002	.001
JP4	.059	.001	.407	.000	.001
JP3	.058	.001	.428	.000	.001
JP2	.052	.001	.405	.002	.001
JP1	.061	.001	.494	.001	.001
KA6	.063	.001	.539	.000	.001
KA4	.061	.001	.549	.001	.001
KA3	.084	.001	.434	.000	.001
KA2	.075	.001	.540	-.001	.001

Scalar Estimates (Group number 1 - Default model)

Regression Weights: (Group number 1 - Default model)

Parameter	Estimate	Lower	Upper	P
KA <--- TE	.644	.479	.839	.000
KT <--- TE	.513	.300	.788	.000
KT <--- KA	.294	.049	.564	.021
JP <--- TE	.449	.212	.721	.001
JP <--- KT	.216	.051	.392	.010
JP <--- KA	.441	.215	.723	.000
TE1 <--- TE	1.000	1.000	1.000	...
TE2 <--- TE	1.043	.864	1.251	.000
TE3 <--- TE	1.044	.860	1.268	.000
TE4 <--- TE	1.133	.951	1.354	.000
TE5 <--- TE	1.018	.836	1.253	.000
TE6 <--- TE	1.060	.854	1.315	.000
TE8 <--- TE	1.016	.839	1.240	.000
KT1 <--- KT	1.000	1.000	1.000	...
KT2 <--- KT	.869	.724	1.044	.000
KT3 <--- KT	.884	.738	1.069	.000
KT4 <--- KT	.964	.800	1.166	.000
KT5 <--- KT	1.203	1.022	1.437	.000
KT6 <--- KT	1.071	.874	1.311	.000
KT7 <--- KT	1.068	.891	1.300	.000
KA1 <--- KA	1.000	1.000	1.000	...
KA2 <--- KA	1.248	1.010	1.531	.000
KA3 <--- KA	1.115	.887	1.405	.000
KA4 <--- KA	1.192	.977	1.486	.000
KA6 <--- KA	1.148	.970	1.376	.000
JP1 <--- JP	1.000	1.000	1.000	...
JP2 <--- JP	.874	.712	1.075	.000
JP3 <--- JP	.962	.797	1.161	.000
JP4 <--- JP	.983	.785	1.233	.000
JP5 <--- JP	.881	.724	1.081	.000
TE7 <--- TE	1.114	.921	1.352	.000

Standardized Regression Weights: (Group number 1 - Default model)

Parameter	Estimate	Lower	Upper	P
KA <--- TE	.669	.572	.752	.000
KT <--- TE	.459	.280	.632	.000
KT <--- KA	.253	.036	.450	.023
JP <--- TE	.399	.196	.612	.001
JP <--- KT	.214	.045	.369	.011
JP <--- KA	.377	.192	.571	.000
TE1 <--- TE	.763	.692	.817	.001
TE2 <--- TE	.731	.662	.788	.000
TE3 <--- TE	.722	.655	.778	.000
TE4 <--- TE	.764	.699	.817	.000
TE5 <--- TE	.705	.623	.767	.001
TE6 <--- TE	.707	.618	.775	.001
TE8 <--- TE	.696	.601	.770	.000
KT1 <--- KT	.719	.626	.792	.001
KT2 <--- KT	.743	.673	.799	.001

e28	.228	.160	.335	.000
e27	.103	.065	.165	.000
e1	.229	.188	.297	.000
e2	.303	.247	.369	.000
e3	.321	.264	.394	.000
e4	.294	.238	.367	.000
e5	.336	.269	.428	.000
e6	.360	.282	.471	.000
e7	.259	.202	.339	.000
e8	.352	.276	.456	.000
e9	.373	.290	.488	.000
e10	.245	.203	.299	.000
e11	.275	.219	.340	.000
e12	.457	.343	.597	.000
e13	.282	.217	.369	.000
e14	.371	.285	.489	.000
e15	.334	.263	.429	.000
e16	.356	.284	.452	.000
e17	.393	.283	.538	.000
e18	.479	.334	.643	.000
e19	.347	.255	.473	.000
e20	.334	.256	.449	.000
e21	.417	.313	.551	.000
e22	.461	.359	.576	.000
e23	.504	.386	.646	.000
e24	.572	.447	.726	.000
e25	.406	.321	.511	.000

Squared Multiple Correlations: (Group number 1 - Default model)

Parameter	Estimate	Lower	Upper	P
KA	.448	.328	.565	.000
KT	.430	.318	.538	.001
JP	.746	.631	.832	.002
JP5	.437	.327	.548	.001
JP4	.407	.293	.521	.000
JP3	.427	.315	.539	.000
JP2	.402	.299	.502	.001
JP1	.493	.366	.608	.000
KA6	.539	.411	.657	.000
KA4	.548	.417	.657	.001
KA3	.435	.274	.603	.000
KA2	.540	.387	.679	.000
KA1	.454	.330	.574	.000
KT7	.577	.465	.676	.001
KT6	.552	.436	.649	.001
KT5	.672	.574	.752	.001
KT4	.448	.315	.579	.000
KT3	.532	.427	.626	.000
KT2	.552	.453	.638	.001
KT1	.518	.392	.628	.001
TE8	.484	.361	.593	.000
TE7	.605	.517	.685	.001
TE6	.500	.382	.600	.001
TE5	.497	.388	.588	.001

Summary of Bootstrap Iterations (Default model)

(Default model)

Iterations	Method 0	Method 1	Method 2
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
5	0	0	0
6	0	2	0
7	0	139	0
8	0	813	0
9	0	1481	0
10	0	1114	0
11	0	843	0
12	0	358	0
13	0	157	0
14	0	70	0
15	0	17	0
16	0	4	0
17	0	2	0
18	0	0	0
19	0	0	0
Total	0	5000	0

0 bootstrap samples were unused because of a singular covariance matrix.
0 bootstrap samples were unused because a solution was not found.
5000 usable bootstrap samples were obtained.

Bootstrap Distributions (Default model)

ML discrepancy (implied vs sample) (Default model)

	505.728	-----
	541.825	*
	577.922	****
	614.019	*****
	650.115	*****
	686.212	*****
	722.309	*****
N = 5000	758.406	*****
Mean = 701.785	794.502	*****
S. e. = .944	830.599	***
	866.696	**
	902.792	*
	938.889	*
	974.986	*
	1011.083	*

ML discrepancy (implied vs pop) (Default model)

	423.950	-----
	435.705	**

	-15.935	*****
	100.339	*****
	216.614	*****
N = 5000	332.888	*****
Mean = 190.428	449.162	*****
S. e. = 3.013	565.436	*****
	681.710	**
	797.984	*
	914.258	*
	1030.532	
	1146.807	*

K-L overoptimism (stabilized) (Default model)

	-21.309	*
	12.520	*
	46.349	*
	80.178	****
	114.007	*****
	147.837	*****
	181.666	*****
N = 5000	215.495	*****
Mean = 193.155	249.324	*****
S. e. = .887	283.153	*****
	316.982	****
	350.811	**
	384.640	*
	418.469	*
	452.299	*

Model Fit Summary

CMIN

Model	NPAR	CMIN	DF	P	CMIN/DF
Default model	56	381.537	269	.000	1.418
Saturated model	325	.000	0		
Independence model	25	3223.628	300	.000	10.745

RMR, GFI

Model	RMR	GFI	AGFI	PGFI
Default model	.032	.889	.866	.736
Saturated model	.000	1.000		
Independence model	.279	.212	.146	.196

Baseline Comparisons

Model	NFI Delta1	RFI rho1	IFI Delta2	TLI rho2	CFI
Default model	.882	.868	.962	.957	.962
Saturated model	1.000		1.000		1.000
Independence model	.000	.000	.000	.000	.000

RMSEA

Model	RMSEA	LO 90	HI 90	PCLOSE
Default model	.042	.032	.051	.924
Independence model	.202	.196	.208	.000

AIC

Model	AIC	BCC	BIC	CAIC
Default model	493.537	507.209	688.453	744.453
Saturated model	650.000	729.343	1781.208	2106.208
Independence model	3273.628	3279.732	3360.644	3385.644

ECVI

Model	ECVI	LO 90	HI 90	MECVI
Default model	2.065	1.865	2.298	2.122
Saturated model	2.720	2.720	2.720	3.052
Independence model	13.697	12.949	14.476	13.723

HOELTER

Model	HOELTER .05	HOELTER .01
Default model	194	205
Independence model	26	27

Execution time summary

Minimization:	.063
Miscellaneous:	.760
Bootstrap:	6.739
Total:	7.562